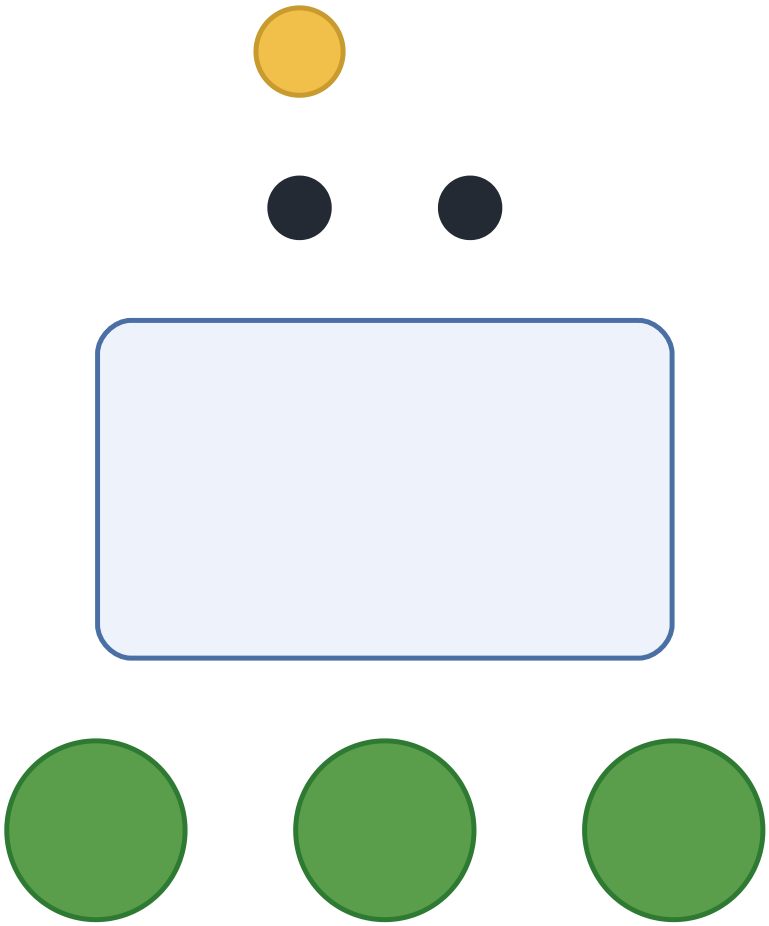
Apitor Remote

The big manual for young pilots

robots Robot R · J · S · Q · X · Wheels

for kids 6–15 and their grown-ups



source: docs/instructions-en.md · Russian version: instructions-ru.pdf

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How to read this manual. Keys are typed on the keyboard, so they always sit in a box: `q`, `w`, `Shift+Q`. Programs can be written with words — like this:

```
repeat 4
forward 4
right 2
```

A yellow box is a hint, a red one says “careful”, a green one is a challenge: try it yourself. Notes like “section 5, page 7” point to other pages. The ⋆ marks show how hard a task is: ⋆ easy, ⋆⋆ think a little, ⋆⋆⋆ for masters.

Where to start:

If you want to…	Open
just drive around	sections 3–5
teach the robot a program	sections 6–8
put on a show with light	sections 9 and 10
send the robot back along its tracks	section 12
fix something that went wrong	section 13
install and tune the remote (grown-ups)	section 14

1. What the remote is

The “remote” is a program on your computer. It talks to the Apitor robot over Bluetooth and tells it what to do. You press keys — the robot drives, spins and blinks its light.

What you need:

- an **Apitor robot** — built and with a charged battery;
- a **computer with Bluetooth** — a laptop, or a desktop with a Bluetooth adapter;
- a **terminal window** where the remote is already running.

When everything is ready, the screen is arranged like this: the last program is in a frame at the top, the big keys `W A S D` with arrows are in the middle, and the diary is in a frame at the bottom. The diary shows something like this:

```
[info] Apitor remote — Robot R (product 'r')
[robot] looking for the robot...
```

The remote is not connected to the robot yet — that happens when you press `q` (section 3, page 5). If there is no welcome line at all, ask a grown-up: section 14, page 18.

What the remote can do:

What	How	Section
drive forward and back, spin around	`w` `s` `a` `d`, arrows	4
change the speed, switch the boost on	`]` `[`, `Shift`	4
light the LED up, blink, switch it off	`1` … `8`, `0`, `b`	5
spin motor 3 (an accessory)	`v` `V`	2
drive a program made of steps	`p`, then `Enter`	6–8
record a trip and drive it back	`m`, `r`, `t`	12
keep programs in files	`f`, `g`, `e`	11

What the remote cannot do: it does not drive the robot from far away (stay within a few metres), it does not replace the official app, and it cannot carry the robot up the stairs.

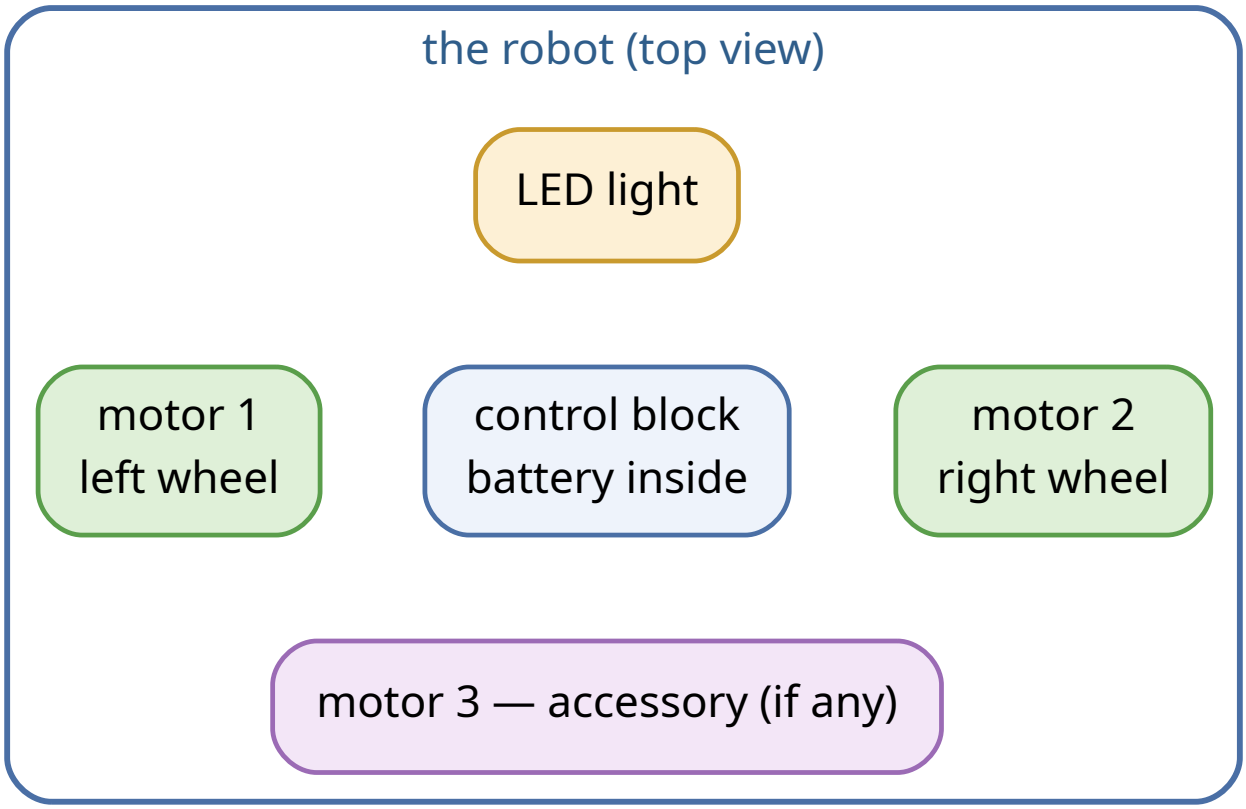
The remote also tells you what is going on: the bottom frame is its diary with the lines `[robot]`, `[prog]`, `[run]`. They show what the robot is doing right now:

```
[robot] connected: Robot R (AA:BB:CC:DD:EE:FF)
[prog] ↑5 · →2 (2 steps · ~1.8 s · speed 6)
[run] done: 2 steps in 1.8 s (planned 1.8 s)
```

Robots come in different shapes: `Robot R`, `J`, `S`, `Q`, `X` are cars, `Wheels` is a wheeled base. Inside they are alike: two wheel motors, an LED light and a third motor for accessories — so the remote works with every model.

> **Challenge “First look”.** Open the remote and find three parts of the screen: the last-program frame, the big `W A S D` keys with arrows, and the diary frame at the bottom. Then find the `h` key — the remote will tell you about the keys itself.

2. Meet the robot



The robot seen from above: two wheels on the sides, the control block in the middle, the LED light in front, motor 3 for an accessory

- **Motor 1 and motor 2** — the left and the right wheel. When the wheels spin in opposite directions, the robot turns on the spot.
- **Control block** — the robot's "brain" and the battery. If the robot stays silent, the battery probably needs charging.
- **LED light** — a small lamp you can control from the keyboard (section 5, page 7).
- **Motor 3** — for extra mechanisms, if your model has them. Key `v` turns it forward, `V` turns it backward (it is a switch: press once to start, press again to stop).

Here are the same parts in one table:

Robot part	What it is	Keys
Motor 1 and motor 2	the left and the right wheel	`w` `s` `a` `d`
LED light	the lamp in front, seen from afar	`1` … `8`, `0`, `b`
Motor 3	an axle for accessories (if you have them)	`v` `V`
Control block	the "brain" and the battery	—

The robot drives "relative to itself": `w` goes wherever its nose points. Turn the robot nose away from you and everything becomes obvious.

> **Challenge "Robot exam".** Point at motor 1, motor 2, motor 3, the LED light and the control block. Remember which key turns each one on.

3. The first connection

1. Switch the robot on.
2. Put it on the floor — away from table edges, feet and pets.
3. Press `q`. The remote starts looking for the robot.
4. Wait a few seconds — you will see something like:

```
[robot] looking for the robot (powered on and not connected to a phone?)...
[robot] found Robot R (AA:BB:CC:DD:EE:FF), connecting...
[ok] connected: Robot R (AA:BB:CC:DD:EE:FF)
```

5. All set! The keys drive the robot now.

If the robot is connected to a phone, the computer cannot find it: Bluetooth talks to one device at a time. Close the phone app and press `q` again.

Next time `q` connects almost instantly: the robot's address is remembered. If the robot is switched off, the remote scans again and tells you about it.

If something is wrong	What to do
It scans and never finds the robot	is the robot on? is the battery charged?
A phone has taken the robot	close the phone app and press `q` again
The wrong robot was found	`Shift+Q` and start with `--product` (section 14)
It is connected but will not drive	press `w` while the remote window is active; check `]` (speed)
The robot hums without moving	lower the speed: `[`
Want to check who is connected	the first line of the window: `[robot] connected: …`

To disconnect, press `Shift+Q`. That is handy when the game is over: the robot stops and stops waiting for commands.

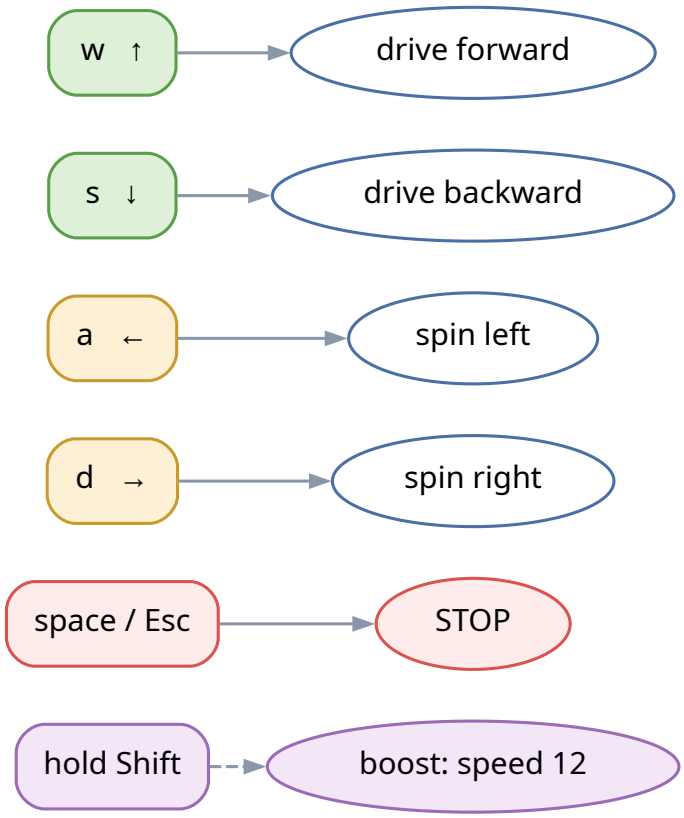
Challenge "I can connect myself". Switch the robot on and connect with `q` without help. Then disconnect (`Shift+Q`) and connect again: the second time is faster, because the robot's address is remembered.

Why is the second time faster? The remote remembers the robot's address in the file `addresses.json` (section 14, page 18), so `q` connects right away, without scanning. If the robot does not answer, the remote starts scanning again and refreshes the address.

If there are several robots around (or a friend has the same one), tell the remote which model you need: `--product r`, `--product j` — and so on up to `--product w`. And `--no-cache` makes it scan from scratch every time.

How long to wait? Usually two or three seconds. If fifteen have passed and there is no robot yet, press `q` again: the remote will start a new search. The battery lasts a long time, but before a long game it is best to put the robot on charge.

4. How to drive



Driving keys: w — forward, s — backward, a — left, d — right, space — stop, Shift — boost

Keys	What they do
`w` or `↑`	forward
`s` or `↓`	backward
`a` or `←`	spin left on the spot
`d` or `→`	spin right on the spot
`space` or `Esc`	STOP — the robot freezes instantly
`]` `[` or `+` `-`	speed: 1 (very slow) … 12 (fast)
`Shift` (hold)	boost: speed 12 while held, the old speed comes back after
`j` or `k`	how to drive: “hold” or “cruise control”

There are two ways to drive, switched with `j` and `k`:

- **hold**: the robot drives while the key is held down. Let go — it stops. This is the best way to learn and to park.
- **cruise control**: like in a car. Press `w` once and the robot drives on by itself, hands free; `space` is the brake — press it and the robot stops. Great for long straights and one-step programs.

> The `Shift` boost works only when the remote can watch real key presses (on Linux that needs a permission, see section 14, page 18). If boost does not come on, just drive at the usual speed. And remember: `Shift+Q` disconnects the robot — that is not the boost.

> **Challenge “Warm-up”.** Put the robot in the middle of the room. Drive forward for two counts, stop, come back. Then spin a quarter turn (`a` or `d`). Done? You are ready for section 7, page 9.

5. The light



Light colours: keys 1–8, key 0 switches the light off

Key	Colour	Key	Colour
`1`	red	`5`	cyan
`2`	orange	`6`	blue
`3`	yellow	`7`	purple
`4`	green	`8`	white

- `0` — switch the light off entirely.
- `b` or `B` — blink: the LED flashes quickly in the current colour. The colour itself does not change.

Colours can be written into a program as well: `color 1` is red, `color 5` is cyan, `color 0` switches the light off. In program mode the key `c` walks through the colours, and `c` plus a digit picks an exact one: `c3` is yellow.

> **Game “Traffic light”.** One player is the traffic light: yellow, red, green. The second player is the driver: stop on red, drive on green, get ready on yellow. Then swap.

> **Challenge “Garland”.** In program mode collect: `color 1`, `space`, `color 4`, `space`, `color 3`, `space` — and press `Enter`. The robot stands still and blinks three colours, like a garland on a tree.

For example, this little traffic-light program:

```
color 1
pause 1
color 4
pause 1
color 0
```

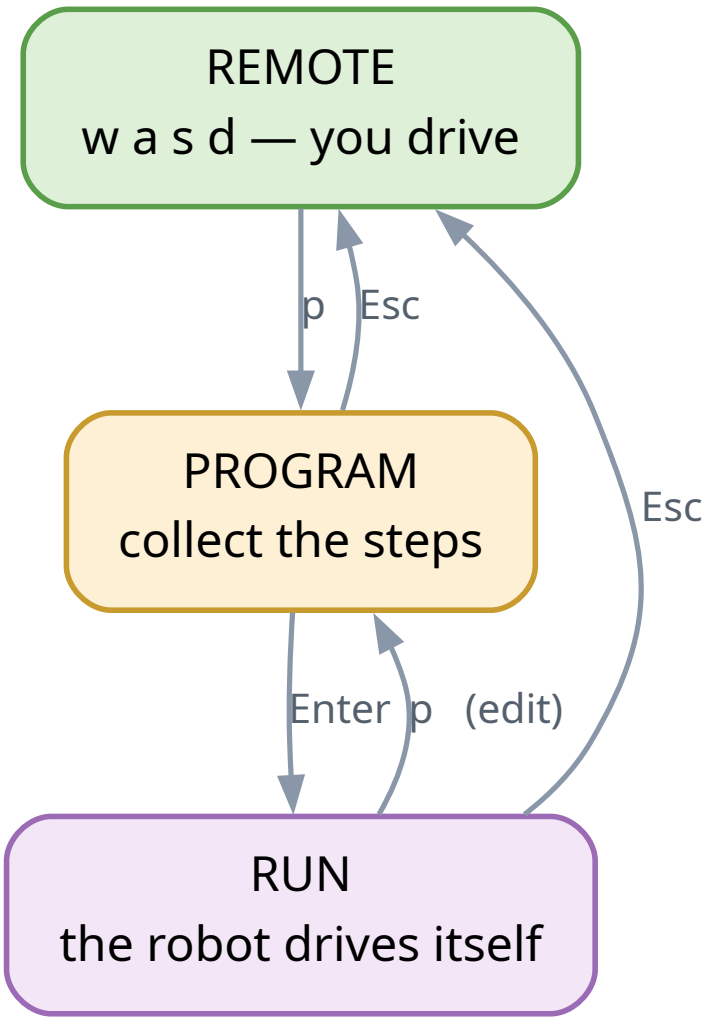
The robot stands in red, then in green, and then switches the light off — and you can drive on. The colour stays on until you press another one, just like a real toy car with a flashing light.

Where to press what — to avoid confusion:

I want to	In the remote	In a program
light a colour	digits `1` … `8`	the step `color 4` or `c4`
switch the light off	`0`	the step `color 0`
blink	`b` or `B`	the step `blink` (or `b`)
walk through the colours	—	the `c` key

> A blink does not change the colour: `b` flashes whatever is on right now. And digits in program mode do not light anything up — they land in a step: `c3` means “light up yellow”, not “light up now”.

6. Three remote modes



The three modes: remote, program, run — and the keys that move you between them

- **Remote** — the keys drive the robot (sections 4 and 5, pages 6 and 7).
- **Program** — you collect the steps; the robot stands still and waits.
- **Run** — the robot performs the program by itself.

How to move around:

Keys	Where you get
`p` or `P`	into the program mode — always, from anywhere
`Enter`	start the program (the run begins)
`Esc`	back to the remote (the program stays in memory)
`Enter`	when paused: continue from here
`r` `t`	in the remote: drive the route back (section 12, page 14)

A trick: if you press `p` while the robot is driving, the remote stops it and opens the program. You can leave the program mode only with `Esc` — pressing `p` inside the program does nothing. No accidental way out.

During a run: `space` pauses and resumes, `Esc` stops and returns to the remote, `p` stops and goes straight to editing the program.

The LED shows the state: one green flash — start, two green flashes — finish, red — the program was interrupted. After the finish the previous colour comes back.

7. Your first program

A program is a list of steps: drive, turn, wait, blink, change the colour. You collect them in the program mode (`p`), or once — `o`.

Keys	A program step
`w` `s` `a` `d` (or arrows)	forward, backward, left, right
digits `1`…`9` `0`	how many times: `wwwww` = "forward 5", `w5` as well, `0` means ten
`space`	pause
`b`	blink
`c`	next colour; `c` plus a digit picks an exact colour (`c3` — yellow)
`]` `[`	program speed (digits right after: `]12`)
`*` (star, `Shift`+`8`)	repeat the whole program: `*` cycles 1→2→…, `*5` runs it 5 times
`Backspace`	delete the last step
`x`	clear the whole program
`Enter`	start it
`Esc`	back to the remote

The keys `r` `R` `t` `T` work in this mode too: the robot drives the route back, and the program stays where it is (section 12).

Under the prompt the remote shows your program as symbols:

```
[prog] ↑5 · →1 · ✦2 (3 steps · ~4.2 s · speed 6)
```

The same symbols are used in files and in these diagrams:

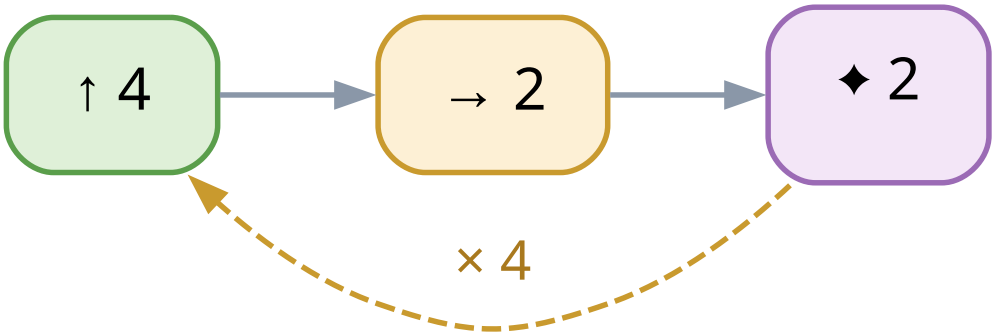
Symbol	Step	Word
↑	forward	`forward` or `вперёд`
↓	backward	`back` or `назад`
← →	left, right	`left` `right`
‖	pause	`pause` or `пауза`
✦	blink	`blink` or `мигнуть`
●	colour	`color 3` or `цвет 3`

The number next to a symbol is how many times the step repeats.

How long it takes. One "forward" step is about 0.6 seconds (the number is multiplied by 0.6), a turn is the same, a pause is 1 second and one blink is 0.25 seconds. The remote shows an estimate right away (`~4.2 s`) and the real time after the run.

> **Challenge "Knock-knock".** Program: `forward 2`, `pause 1`, `back 2`. What happened? What if the pause becomes a blink — can the robot "knock" with light?

8. The “Square” program



Square: forward, turn, blink — four times around

The square is the first real program. The robot drives forward, turns, and repeats that four times — and you get a square. Collect it in the program mode:

```
w4      forward 4
d2      spin right 2
b2      blink 2
*4      repeat everything four times
```

Press `Enter` — and away it goes. The key `*` (the star) repeats the whole program, and the digit after it says how many times: `*4` means four laps. The very same program can be written with words in a program file (section 11, page 13):

```
repeat 4
forward 4
right 2
blink 2
```

A quarter turn is about `d2`. If your square comes out lopsided, try `d1` or `d3`: on different floors the robot turns a little differently. That is normal — the robot does not know whether it is on carpet or tiles.

Challenge “Big square”. Make a square twice as large: `forward 8`, `right 2`, `repeat 4`. Guess how many seconds the run will take and check yourself against the `[prog]` line.

Why is it a square? The turn `right 2` is about a quarter circle, 90°. Four turns together make 360°, a full circle, so the robot comes back to where it started. Put `right 1` and the robot turns only an eighth — and the square turns into a crooked heptagon. Maths!

Challenge “Rectangle”. Make two sides long and two short: `forward 8`, `right 2`, `forward 4`, `right 2` and `*2` at the end. Explain why repeating twice was enough.

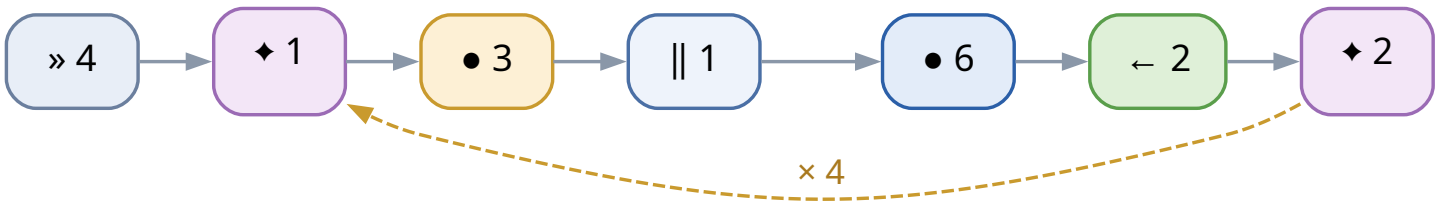
The `[prog]` line shows the length of the program: `(3 steps · ~4.2 s)`. So four laps take about sixteen seconds — just enough time to count to sixteen and check on the robot.

9. The “Firefly” program

Now a dance with light: the robot spins while the LED changes colour. Collect the program (or write it in a file and load it with `g`):

```
speed 4
repeat 4
blink 1
color 3
pause 1
color 6
pause 1
left 2
blink 2
```

What happens: the robot blinks yellow, waits, flashes blue, waits, spins left and blinks once more — four times around. A little show.



The “Firefly” scenario: blink, yellow, pause, blue, turn, blink — and four laps

Step	What the robot does
`speed 4`	drives slower: the dance comes out smooth
`repeat 4`	runs the whole program four times
`blink 1`	flashes in the current colour
`color 3`	lights up yellow
`pause 1`	stands for a second
`color 6`	lights up blue
`left 2`	spins on the spot
`blink 2`	blinks twice

Challenge “Your own firefly”. Change the colours and pauses to your own and make a show of four laps. Try `speed 8` and then `speed 2` — the dance comes out completely different.

A colour change is a step too: the robot changes the light not immediately, but when it reaches that step. That is why a “dance” of colours and pauses looks like a real light show.

10. Challenges

Stars show the difficulty: ⋆ — easy, ⋆⋆ — think a little, ⋆⋆⋆ — for masters.

1. ⋆ **The straight line.** Drive from one edge of the carpet to the other and come back.
2. ⋆ **Traffic light.** Program: red, pause, green, blink, drive.
3. ⋆⋆ **The snake.** Forward, turn right, forward, turn left — three times.
4. ⋆⋆ **Two squares.** The square from section 8 (page 10) with `repeat 2` — the robot must come back to where it started.
5. ⋆⋆ **The dance.** Spin, blink and change colours so that the grown-ups can guess the tune.
6. ⋆⋆⋆ **Boost race.** A long straight at speed 12 (the `Shift` boost) — without hitting the target.
7. ⋆⋆⋆ **The staircase.** Forward, right, forward, right… Build a path of steps to your toy. How many steps does it take?

Hints (open them only if you are really stuck):

8. ⋆⋆ **The figure of eight.** Two circles in opposite directions — an infinity sign.
9. ⋆⋆ **The patrol.** Drive around the room and come back to the same spot with the key `r`.
10. ⋆⋆⋆ **The repeater.** Drive around, record the trip as a macro (`m`, stop with `Esc`) and send the robot home with `r`.

Hints (open them only if you are really stuck):

- The snake: `w3 d1 w3 a1` — and repeat a few times.
- Two squares: put `repeat 2` at the very beginning of the program.
- The staircase: tune the turn number until you get a neat quarter turn (start with `d2`).
- The race: accelerate on the straight and press `space` before the target.
- The figure of eight: a circle left (`w2 a2`), a circle right (`w2 d2`) — and repeat.
- The patrol: drive back the way you came — or just press `r`.

Challenge diary. Write down what worked (and when):

No.	Challenge	Done
1	The straight line	☐
2	Traffic light	☐
3	The snake	☐
4	Two squares	☐
5	The dance	☐
6	The boosted race	☐
7	The staircase	☐
8	The figure of eight	☐
9	The patrol	☐
10	The repeater	☐

11. Program files and the editor

You can save a program into a file and load it next time.

Keys	Action (in the program mode)
`f`	save: the remote asks for a name, `Enter` accepts it
`g`	load: a list of files, type a number and `Enter`
`e` (`E`)	edit the program in a full-size editor (works from the remote too)
`x`	clear the program

Program files live in the `programs/` folder next to the remote. You can open them in a text editor and write with words — in English or in Russian:

```
# square: four sides and a blink at the end
speed 5
repeat 4
forward 4
right 2
blink 2
```

Every word the remote understands:

Word	Symbol	Meaning
`forward`, `вперёд`, `up`	`↑`	drive forward
`back`, `назад`, `down`	`↓`	drive backward
`left`, `влево`	`←`	turn left
`right`, `вправо`, `rght`	`→`	turn right
`pause`, `пауза`	`‖`	stand still
`blink`, `мигнуть`	`✦`	blink
`color 3`, `цвет 3`	`●3`	change the colour
`repeat 4`, `повтор 4`	—	run everything four times
`speed 5`, `скорость 5`	`»5`	change the speed from here
`forward 1.2s`, `вперёд 1.2с`	`↑1.2s`	drive for exactly 1.2 seconds

Anything after `#` is a comment: the remote does not read it. Big and small letters, `ё` and `е`, commas and `×` are all forgiven. And it understands English and Russian words at once.

A file loads into the one-time program the same way: `o`, then `g`. The remote says “loaded one-time” and leaves the `p` program alone.

Challenge “Out and back by file”. Collect the square, save it with `f` as `square2.txt`, clear it (`x`) and load it again with `g`. Then press `e` and add the line `color 5` at the very top.

The program you collected will not be lost even if you close the remote: it keeps a draft and picks it up next time. Ready-made examples live in the `examples/` folder.

12. The macro and the way back

You do not have to collect a program by hand — you can **show** it: let the remote record your trip. Such a recording is called a **macro**.

How to record:

1. Press `m` — the log shows `[rec] recording a macro`.
2. Drive the robot as usual: `w` `a` `s` `d`, the light, the turns, the pauses. The remote records everything, and every ride becomes a step with seconds.
3. Press `Esc` (or `n`) — the recording is ready and already sits in the program line. `m` only starts the recording: pressing it again just reminds you that the recording is on.

```
[rec] recording a macro: driving, light, speed are all recorded, pauses become waits; Esc or n — stop
[rec] ↑
[rec] ←
[rec] macro recorded: 3 steps · 2.88 s — Enter — drive, r/t — back, f — save, p — edit
```

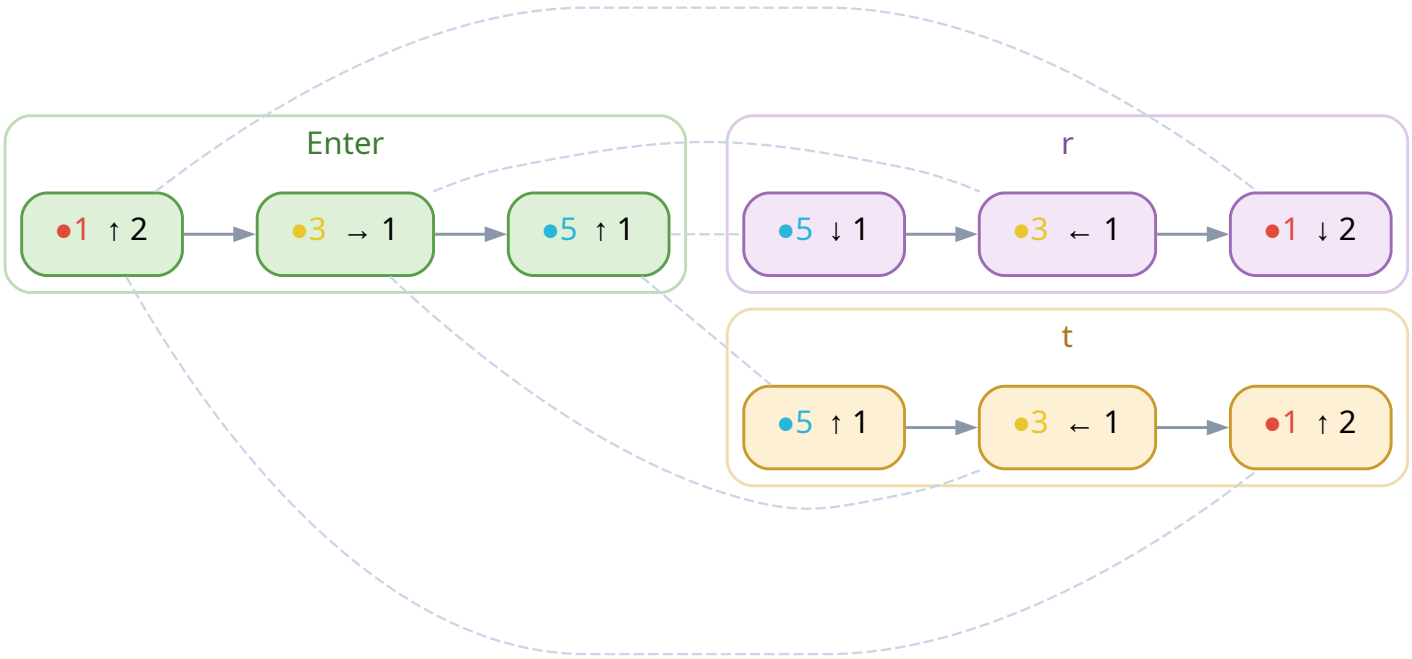
Now `Enter` — and the robot repeats the trip by itself: with the same pauses, light and speed. A macro is an ordinary program: you can edit it (`p`), save it into a file (`f`) and load it next time (`g`).

> While the recording is on, `p`, `Enter`, `r`, `t`, `x`, `f`, `g` do not work: finish the recording first — `Esc` or `n`. And `space` is just a robot stop: the recording continues.

The way back: `r` and `t`

The robot drove away to the other end of the room. How to bring it back along its own tracks without collecting a second program? There are two keys for that:

- `r` — **backwards**. The robot drives the route from the end, and everything is mirrored: “forward” becomes “backward”, “left” becomes “right”. It backs along its own tracks.
- `t` — **the robot was turned by hand**. At the end of the route you turned it 180°, so it drives back forward: only the turns change (left ↔ right).



The way back: the same route from the end — backwards and forward

The duration is kept: if the robot drove for 2 seconds on the way out, it drives for 2 seconds on the way back as well (numbers like `↑5` become seconds).

The speed and the colour switch exactly where you changed them. You drive at 6 first, then switch to 12 and finish the route — on the way back the robot starts at 12 right away (that stretch was driven at 12), and at the turning point it

switches back to 6 and keeps it to the end. The colour works the same way: a stretch driven in cyan is driven back in cyan, and the colour switches exactly when you reach that stretch (in the picture `●5`, `●3`, `●1` stand in a different order). Stretches that ran "with whatever was lit" get the colour that was on before the run.

> `Shift+Enter` drives the program without pauses, and `R` or `T` (`Shift` with `r` or `t`) does the same on the way back. Capitals come by themselves, and `Shift+Enter` is learned from the terminal itself (kitty, foot, alacritty and others) — or from the keyboard tracking, when it is on (a grown-up knows what that is: section 14, page 18).
>
> The way back works from program mode too: type the steps, press `r` (or `t`) — the robot drives, and the program stays where it is.

The way back does not spoil the program: it is built on the fly, so after it `Enter` drives forward again and `p` shows the old steps. And `repeat` has nothing to do with it: back the robot always drives once — after the first pass it already stands at the start of the route and looks the other way.

> **Challenge "Out and back".** Drive a route with two speeds and two colours, record it as a macro and come back with `r`. Look at the `[run]` lines: you can see the robot started the way back at the last speed.

All four keys side by side:

Keys	What the robot does
`r`	drives the route back, backwards
`t`	the same, but forward — for a robot you turned by hand
`R` `T`	the same without pauses, in one go
`space`, then `r` / `t`	drive back only the part already done
`space`, then `Enter`	return to the normal route and finish it from here

In the remote window you can see the speed and the steps of the way back:

```
[run] start: 9 steps · speed 6 · reversed (driving backwards)
[run] ▶ »12
[run] ▶ ←0.6s
[run] done: 9 steps in 4.6 s
```

While the recording is running the window shows lines like these:

```
[rec] recording a macro: driving, light, speed are all recorded,
      pauses become waits; Esc or n — stop
[rec] … do something (Esc or n — stop)
```

And when you stop the recording — how many steps you collected:

```
[rec] macro recorded: 7 steps · 5.4 s — Enter — drive, r/t — back,
      f — save, p — edit
```

What to try with the way back:

- The letter "U": forward, turn, forward, turn, forward — then press `r` and the robot drives the letter backwards by itself.
- A traffic-light track: write `color 1` at the start and `color 5` at the end of the route — on the way back the colours come in a different order.
- A long trip: write `speed 12` in the middle of the file — on the way back the robot drives fast first and comes home slowly.
- A return halfway: start the "long trip" and press `space` when the robot has driven only part of the route, then `r` — it comes home the short way, only over what was done.

The robot went the wrong way: drive back what was done

Sometimes the robot goes the wrong way. You do not have to wait for the end of the program: pause the run (`space`) and press `r` or `t` — the robot drives back **only along the part of the route it has already driven**. This is the quickest way to bring it back.

If the pause caught the robot in the middle of a long step, it drives back exactly as many seconds as it managed to drive. The program "forward 20s", paused at the 16th second — the way back starts with "back 16s", and then come the steps already driven, in reverse order. Everything else is like a usual way back: the turns are mirrored, the pauses and the light stay in place, and `R` and `T` — without pauses. The program itself does not change: `Enter` after the return drives it forward again.

```
[run] paused at step 1 of 1 — driving back what was done
[run] start: 1 step · speed 6 · reversed: the part already done (driving backwards)
[run] ▶ ↓16s
```

Without a pause `r` and `t` do nothing during a run: the remote hints that you should press `space` first.

A pause spoils nothing: press `space` again and the robot calmly drives on from the same place. `r` or `t` end the run and start the way back. So while paused you have a choice: drive on or come home.

If the program repeated, the way back runs over everything driven: the whole first pass and the part of the second one the robot managed to drive. It comes back to where the program started — the same place a full way back ends at.

Challenge "Stop, wrong way!". Start a long program and press `space` when the robot has driven only part of the route. Then press `r` — and look at the `[run]` lines: the number of seconds it drove back is exactly how far it had got.

What goes into the way back:

What was there	What happens on the way back
a step already driven	mirrored: `forward` ↔ `backward`, `left` ↔ `right` (with `t` only the turns mirror)
a long step in flight	a stretch in seconds: as far as it got
a pause already passed	a pause, as in a full way back (with `R` and `T` — without pauses)
a blink	a blink
light and speed	they switch where you changed them

If the way back finishes on its own, it does not drive the rest of the program: the run is over and the robot stands where it started. Want the whole program — press `Enter` and it drives from the very beginning (the steps never change). But if you pause the way back and press `Enter`, the robot returns to the normal route and continues it from here — see the next section.

If it was a one-time program (`o`), it is already gone after the return: `Enter` and `r`/`t` use the `p` program again.

Back to the route: `Enter` when paused

If the way back has already started and you change your mind, you do not have to finish it. Pause it (`space`) and press `Enter`: the remote first drives back what the way back has done, and then continues the normal program from where you stopped. `Shift+Enter` does the same without pauses.

Example: "forward 5s, right 5s, forward 20s". Pause at the 16th second of the last "forward", then press `r`: the way back drives 16 seconds backwards and 3 seconds of a left turn. Pause again, then press `Enter` — the robot turns right for 3 seconds and drives forward for 20 seconds: the remote glues the 16 seconds of the return and the remaining 4 into one step.

When a normal run is paused, `Enter` also continues from here, and the program itself never changes.

13. If something is wrong

What happened	What to do
`[err] could not connect`	is the robot on? not busy with a phone? press `q` again
The robot drives backward instead of forward	it is a setting: section 14 (page 18), “calibration”
The robot is too slow or too fast	`]` and `[` change the speed (1…12)
The keys do nothing	the remote listens to the keyboard only while its window is active; press `h` for a hint. Do not type `w a s d` in other windows
`Enter` does not start the program	look at the `[prog]` line: `0 steps` means the program is empty
The robot stopped in the middle of a program	`space` was pressed (pause) — press it again
The robot went the wrong way	`space`, then `r` (or `t`) — the robot drives back only the part already done
Everything is frozen	`Ctrl+C`: the remote stops the motors and quits
Not sure what to press	`h` — help; `Esc` — back to the remote
`[err] the program was not changed`	there is a typo in the file: press `e` and look
A file is missing in the `g` list	check the `programs/` folder (section 14, page 18)
The robot blinks and does not drive	`space` was pressed — press it again
The light is different after a program	that is on purpose: the old colour comes back after the finish
The program ran in a strange way	press `p` and look at the steps; `x` clears them
The robot drove under the sofa	`Esc` — stop; then reach it by hand or reconnect with `q`
The program goes on forever	check `repeat`: the number at the start of the program
The light blinks by itself	that is a blink step (✦) from the program — wait or press `Esc`

First aid, step by step. If the robot behaves strangely:

1. Press `Esc` — the remote stops the robot. This is safe at any moment.
2. Look at the last line in the window: it says what happened.
3. Press `h` for help, `q` to reconnect. The run journal is switched on with the `--log` flag (section 14).
4. If nothing helps at all: `Ctrl+C` and start the remote again.

If the robot has driven far away. Press `Esc` — the remote stops it instantly, wherever it is. Then there are three ways to bring the robot home: drive it yourself (just `w` in the right direction), press `r` (the robot returns along its tracks) or `t` if you have managed to turn it by hand. And if the robot has left the room altogether — `q` to reconnect and `y` to check the motors.

> Almost everything is cured by three things: `Esc`, `q` and `Ctrl+C`. Press them boldly — the robot will not break, and the remote always stops the motors before quitting.

14. For grown-ups: install, flags, tuning

The easiest way: double-click `run.sh` (Linux) or `run.bat` (Windows). The file creates `.venv`, installs the dependency, opens a terminal and starts the remote. The remote opens full-screen right away, and `--no-tui` brings back the plain line-by-line output. Flags are passed through as they are: `run.sh --no-tui` or `run.bat --program examples/square.txt`.

Install by hand (once):

```
cd Apitor-WASD
python3 -m venv .venv
.venv/bin/pip install -r requirements.txt
```

Run:

```
./run.sh                                        # all of that in one command
.venv/bin/python apitor_remote.py               # full-screen remote (default)
.venv/bin/python apitor_remote.py --no-tui      # plain line-by-line output
.venv/bin/python apitor_remote.py --simulate    # no robot: print frames
.venv/bin/python apitor_remote.py --program examples/square.txt
```

On Linux the remote needs permission to read the keyboard (otherwise it warns and falls back to auto-stop): `sudo usermod -aG input "$USER"`, then log out and in again. On Windows Python 3.10+ and Bluetooth are enough.

The boost (hold `Shift` — speed 12) works only when the keyboard tracking is on. `Shift+Enter` and capitals `R`, `T` need no tracking: on start the remote asks the terminal whether it can tell `Shift` apart (the kitty keyboard protocol: kitty, foot, alacritty, wezterm, ghostty, a recent GNOME Terminal) and turns the extended keys on if it can. If you do not need the tracking at all (the auto-stop by `--hold-timeout` is enough), run with `--no-key-state`: the remote will not read the keyboard in `/dev/input` and will not warn.

Useful flags (the full list is in the README):

Flag	Why
`--speed 6`	starting speed 1…12
`--product r`	robot model: `r`, `j`, `s`, `q`, `x`, `w`
`--no-tui`	plain line-by-line output (full-screen is the default; the old `--tui` is still accepted)
`--no-cache`	do not use the saved robot address
`--simulate`	no robot: print the frames that would be sent
`--cruise`	start with cruise control on (the old name `--latched` still works)
`--program-dir DIR`	where program files live (default `programs/`)
`--step-time` `--turn-time` `--pause-time` `--blink-time`	step durations, s (0.6 / 0.6 / 1.0 / 0.25)
`--max-steps N`	limit the program length (0 — no limit)
`--log`	run journal in `~/.config/apitor-control/runs.log`
`--key-state-device PATH`	watch a single keyboard only
`--no-key-state`	do not read the keyboard at all: no permissions, no warnings
`--verbose`	detailed output: protocol frames and scanning

Flag	Why
`--lang ru`	Russian interface

Calibration (if the robot drives the wrong way). Connect and press `y`: the robot drives forward, backward and sideways, so you can see what is mixed up. Then start with the right flag:

```
.venv/bin/python apitor_remote.py --flip-left     # the left wheel spins the wrong way
.venv/bin/python apitor_remote.py --flip-right    # the right wheel
.venv/bin/python apitor_remote.py --left-motor 2 --right-motor 1   # the wheels are swapped
```

Key `i` runs the motors one by one — that shows which port drives which wheel.

This manual. Sources: `docs/instructions-ru.md`, `docs/instructions-en.md`; diagrams: `docs/diagrams/*.dot`. To rebuild the pictures, the HTML and the PDFs:

```
docs/build.sh
```

It needs `pandoc`, `graphviz`, `calibre` and `pdftotext` (poppler) — see `docs/README.md`.

A check without a robot (handy right after the install):

```
.venv/bin/python apitor_remote.py --simulate
```

The remote prints the frames it would send to the robot — you can see what would happen, and no robot is needed.

FAQ.

Question	Answer
The robot is never found	is it on? charged? is a phone holding it? is Bluetooth on?
The remote warns about the keyboard	add the user to the `input` group, or start with `--no-key-state`
Timer stops are not wanted	enable keyboard tracking (the `input` group) or lower `--hold-timeout`
Where is the run journal	`~/.config/apitor-control/runs.log` (flag `--log`)
How to start from scratch	delete the files in `~/.config/apitor-control/` — the remote recreates them
Starting without a terminal	double-click `run.sh` / `run.bat`
Where to take example programs	the `examples/` folder, for instance `examples/square.txt`

All the settings live in the folder `~/.config/apitor-control/`:

File	What is inside
`addresses.json`	robot addresses: so that `q` connects at once
`config.json`	language, step durations, program folder, journal
`runs.log`	the run journal (switched on with `--log`)

Deleting these files is the easiest way to “reset” the remote to factory settings: it recreates them on the next launch.

15. Mini dictionary

- **Remote** — the program that drives the robot from your keyboard (section 1, page 3).
- **Macro** — a recording of your trip (`m`) that becomes a program (section 12, page 14).
- **The way back** — the robot drives the same route from the end (`r`, `t`, section 12, page 14).
- **Step** — one action of a program: drive, turn, blink (section 7, page 9).
- **Program** — a list of steps the robot performs by itself (section 7, page 9).
- **One-time program** — collected with `o`: not saved, lives until the first run and disappears after it (section 7, page 9).
- **Run** — performing the program from start to finish (section 6, page 8).
- **Repeat** — how many times the whole program runs (`*` or `repeat`, section 8, page 10).
- **Hold** — the mode where the robot drives while you hold the key (section 4, page 6).
- **Latched** — the mode where the robot drives until you press `space` (section 4, page 6).
- **Boost** — speeding up to 12 while `Shift` is held (section 14, page 18).
- **Draft** — the program the remote remembers between launches (section 11, page 13).
- **Terminal** — the window where the remote lives; keys are pressed in it (section 1, page 3).
- **Speed** — a number from 1 (very slow) to 12 (fast); `]` and `[` change it (section 4, page 6).
- **The light** — the robot's LED; colours `1`…`8`, and `0` switches it off (section 5, page 7).
- **Motor 3** — the third axle for accessories; `v` starts it, `V` stops it (section 2, page 4).
- **Symbol** — a short way to write a step: `↑5`, `‖`, `✦2`, `●3` (section 7, page 9).
- **Program file** — a text file with steps in the `programs/` folder (section 11, page 13).
- **Comment** — a note after `#`; the remote does not read it (section 11, page 13).
- **Second** — write a number with a letter: `1.2s` instead of "some steps" (section 7, page 9).
- **Challenge** — a task with ⋆ marks; there are ten of them (section 10, page 12).
- **Way back** — the same route from the end (`r`, `t`, section 12, page 14).
- **Recording** — a macro runs until you press `Esc` or `n` (section 12, page 14).
- **Editor** — program mode, where the steps are typed with keys (section 7, page 9).
- **Pause** — a step where the robot stands still and does nothing (section 7, page 9).
- **Cache** — the saved robot address: that is why `q` finds it at once (section 3, page 5).
- **Frame** — a command the remote sends to the robot over Bluetooth (section 14, page 18).
- **Model** — which robot this is: R, J, S, Q, X or Wheels (section 1, page 3).
- **Journal** — the records of runs; switched on with `--log` (section 14, page 18).
- **The square** — the first real program: four turns (section 8, page 10).
- **The firefly** — a dance program: turns and coloured flashes (section 9, page 11).
- **Vocabulary** — every word a program file understands (section 11, page 13).
- **Flag** — a word after the program name on the command line: `--no-tui`, `--simulate` (section 14, page 18).
- **Simulation** — the mode without a robot: frames are printed, nobody is scanned (section 14, page 18).
- **Robot address** — its Bluetooth "name"; the remote remembers it, so `q` is quick (section 3, page 5).
- **STOP** — `space` or `Esc`: the robot freezes and drives no more (section 4, page 6).
- **To blink** — to flash the light briefly (the `b` key; section 5, page 7).
- **Diagram** — a picture in this manual; the sources live in `docs/diagrams/` (section 14, page 18).
- **Cruise control** — the mode where the robot drives on by itself until you press `space` (section 4, page 6).

16. Cheat sheet

Remote:

Keys	What they do
`q` / `Shift+Q`	connect / disconnect
`w` `s` `a` `d`, arrows	drive (forward, backward, left, right)
`space` `Esc`	STOP
`]` `[` `+` `-`	speed up / down
`Shift`	boost (while held)
`1` … `8`, `0`	light colour, `0` turns it off
`b` `B`	blink
`v` `V`	motor 3: forward / backward (a switch)
`p` `P`	program mode
`o`	one-time program: not saved, gone after the run
`Enter`	drive the program: the last macro or a file
`Shift+Enter`	the same, but without pauses
`r` `t`	drive the route back: backwards / turned around
`R` `T`	the same, but without pauses
`m`	record a macro (stop it with `Esc` or `n`)
`f` `g`	save / load a program file
`e` `E`	program editor (works from the remote too)
`y` `i`	test drive / motors one by one
`j` `k`	hold / cruise control
`h` `?`	help
`Ctrl+C`	quit

Program mode: `w/a/s/d` — steps · digits — how many times · `space` — pause · `b` — blink · `c` — colour · `*` — repeat · `]` `[` — speed · `Backspace` — delete a step · `x` — clear · `Enter` — start (with `Shift` — without pauses) · `Esc` — to the remote. **The way back:** `r` `t` — backwards / turned around, `R` `T` — the same without pauses; works from the remote and from the editor. On a paused run — only the part already done.

Run: `space` — pause/resume · `Esc` — stop · `p` — edit. When paused: `r` `t` — drive back what was done. After a run: `Enter` — once more, `r` `t` — back.

A mini example (the square) as a file `programs/square.txt`:

```
speed 5
repeat 4
forward 4
right 2
blink 2
```

In one line (the most needed):

```
q connect · w a s d drive · space STOP · ] [ speed ·
1…8 colour · b blink · p program · o one-time · Enter run ·
r t back · m macro · f g file · e editor · h help · Ctrl+C quit
```

What to write in a program file: `forward 4` · `back 2` · `left 1` · `right 2` · `pause 1` · `blink 2` · `color 3` · `repeat 4` · `speed 5` · `forward 1.5s`. The remote understands Russian words too: `вперёд`, `назад`, `влево`, `вправо`, `пауза`, `мигнуть`, `цвет 3`, `повтор 4`, `скорость 5`. Anything after `#` is a comment.

If you forgot a key. Press `h` — the remote shows the help. The `h` key works in program mode as well: there is a help of its own. And the run journal (the `--log` flag) shows what the robot did and how long it took.

A mini program to memorise (a square with light):

```
speed 5
repeat 4
forward 4
right 2
blink 2
```

How to run it: press `p`, type the steps (or load a file with `g`) and press `Enter`.

Three useful habits:

1. Before a run look at `[prog]`: how many steps and which speed.
2. After a run look at the `[run]` line: it shows how long it took.
3. Point the robot's nose away from you: then `w` means "forward".

Where to go next. The `examples/` folder holds ready programs (the square, the firefly and others) — press `g` and choose one. You can also race a friend: set up skittles and drive the track without knocking any down. The fastest one gets to command the robot!

Three rescue keys:

Key	When it saves you
`Esc`	the robot is doing something wrong — an instant STOP
`q`	the link is lost — connect again
`Ctrl+C`	the remote is frozen — stop the motors and quit

Pilot check:

I can	☐
connect and disconnect (`q`, `Shift+Q`)	☐
drive, turn and change the speed	☐
light any colour and blink	☐
collect a program and run it	☐
record a macro and come back with `r`	☐
save a program and load it	☐
show the remote to a friend	☐

> More programs and examples are in the `examples/` folder. A pilot who has made it this far: build your own track, put down a bottle as a target and try to drive around it without touching it. Good luck!